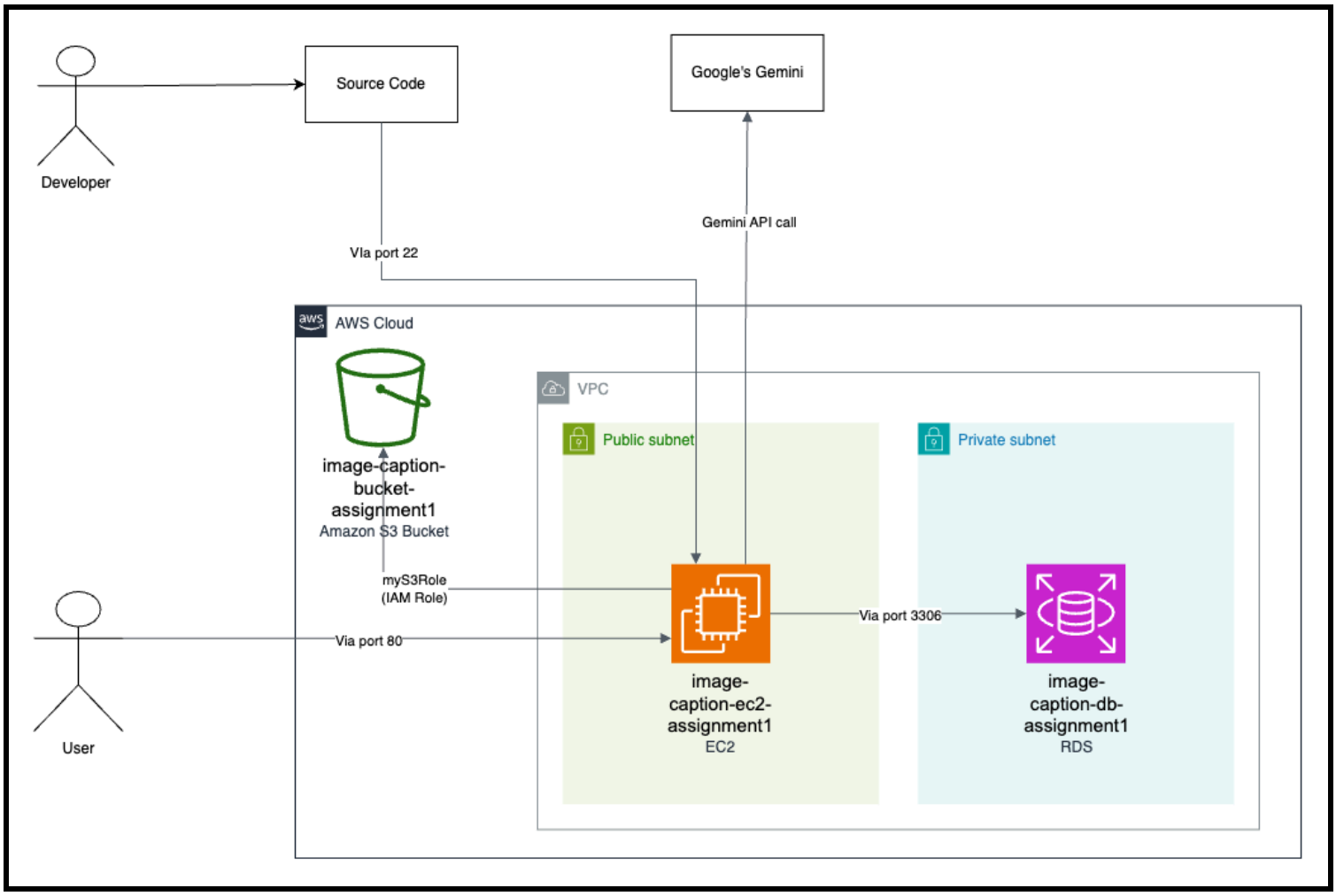


The diagram illustrates the AWS architecture for the Image Captioning Web Application. The diagram has several key components and actors:
The AWS Environment contains the VPC and the Amazon S3 Bucket. The VPC contains a Public subnet in which the EC2 instance is hosted and a Private subnet in which the RDS is hosted. The Developers, Users, Source Code and Google's Gemini are components present in the external environment.

1. Developer: The developer is represented by an actor figure. The developer is responsible for developing and updating the source code.
2. Source Code: The source code box represents the application code, including the app.py and HTML templates. The developer uses SSH protocol to directly deploy the application's source code on the EC2 in the AWS environment via port 22.
3. Google's Gemini: The Google's Gemini box represents Google's Gemini API, the external service used for generating image captions. The EC2 instance from the AWS environment communicates with Gemini via API calls to generate captions for uploaded images.
4. User: The user is represented by an actor figure. The user interacts with the application hosted on EC2 via HTTP protocol on port 80.

5. AWS Cloud Boundary: Represents the AWS cloud environment.
6. Amazon S3 Bucket (image-caption-bucket-assignment1): The green bucket icon represents the Amazon Simple Storage Service (S3) Bucket, which is used for storing images in the AWS environment. The EC2 accesses the S3 with the help of the permissions granted by the IAM role assigned to it. The EC2 stores and retrieves the images uploaded by the user from S3.
7. Virtual Private Cloud (VPC): This is where the EC2 and RDS are hosted.
8. Public Subnet: The green box inside the VPC represents the Public subnet. The EC2 instance is hosted in this subnet, which lets the users communicate directly from the internet.
9. EC2 Instance (image-caption-ec2-assignment1): The orange icon represents the EC2 instance. This is a virtual server in the AWS Cloud environment, where the application's logic and web server components are running. It has an IAM role assigned to it which allows it to communicate with the S3 bucket. The user communicates with the EC2 instance using the HTTP protocol on port 80. It communicates with Google's Gemini via API calls to generate captions for the uploaded images. It communicates with the RDS database via port 3306 to store and retrieve captions generated.
10. Private Subnet: The blue box inside the VPC represents the private subnet. It hosts the RDS database.
11. RDS Database (image-caption-db-assignment1): The purple icon represents the RDS database. This is where the captions generated are stored and retrieved. The EC2 instance communicates to the RDS via port 3306, which is the default port for MySQL.